

TempCheck

Voice-logged food safety — read back and proven. Every flagged temperature is spoken back and confirmed before it's saved, and every number in this deck was measured against AssemblyAI's real API or verified live against the real deployed app — nothing estimated.

THE PROBLEM

Paper logs, wet gloves, one misheard digit

- Every US food establishment is required to log temperatures — most of it is still clipboards, checked with wet, gloved hands, hours after the fact.
- A 38°F reading logged as 48°F isn't a UX nitpick — it's a real food-safety failure, and it's invisible until an inspector (or an outbreak) finds it.
- The fix has to be hands-free (cooks are working) and has to catch the misheard digit *in the same breath*, not at the next audit.

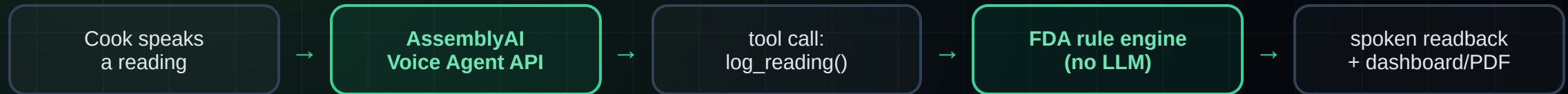
THE IDEA

Say it once. Hear it read back. It's logged.

- A cook calls out a reading mid-task — "walk-in cooler, thirty-eight," "chicken breast, one fifty-two."
- TempCheck logs it, checks it against real FDA limits, and speaks the exact number back before saving anything flagged.
- A cook can self-correct mid-sentence ("steam table, one twenty — no wait, one forty") and TempCheck keeps the corrected number, not the first one.
- Out-of-range readings get a spoken corrective action, not a silent red dot nobody sees until the shift ends.

HOW IT WORKS

Cook → Voice Agent → deterministic verdict → readback



The Voice Agent handles STT, turn-taking, and tool-calling. It never decides whether a reading is safe — that's a separate, unit-tested, deterministic step.

The safety verdict is never a language model's guess

- The model's only job is extraction: location/item, number, unit. A plain, unit-tested comparison against FDA Food Code limits decides safe / amber / red.
- Cold holding $\leq 41^{\circ}\text{F}$ · hot holding $\geq 135^{\circ}\text{F}$ · poultry $\geq 165^{\circ}\text{F}$ · ground/injected meat $\geq 155^{\circ}\text{F}$ · whole muscle $\geq 145^{\circ}\text{F}$ · fish/seafood/eggs $\geq 145^{\circ}\text{F}$ · reheating $\geq 165^{\circ}\text{F}$.
- Plus a two-stage cooling curve ($135 \rightarrow 70^{\circ}\text{F}$ within 2h, then $\rightarrow 41^{\circ}\text{F}$ within 6h, FDA 3-501.14(A)), pairing a spoken "cooling start" with a later "cooling check."
- 39 automated tests cover every category, every boundary, unit conversion, and implausible readings — zero hallucination risk in the verdict itself.

Two real integrations, not one API call

USE	API	WHAT IT DOES
Live cook-facing app	Voice Agent API	Real-time STT + turn detection + tool-calling + TTS, over a live WebSocket session
Accuracy proof harness	Async transcription API (v2)	648 real clips, real upload → transcript → poll, scored against ground truth

We didn't just call the API and hope — we built a real, reproducible test harness against it, published the actual result, and even shipped a no-mic demo mode that replays a sample clip through the exact same live Voice Agent session.

MEASURED ACCURACY — NO INVENTED NUMBERS

570 / 648 correct — 88.0%

92.3%

Baseline noise (299/324)

90.3%

4 accents (130/144)

94.4%

Fast speech, ~1.6x (34/36)

52.8% /

47.2%

Fryer / shouting coworker

Method disclosed *before* the result in `docs/accuracy.md`, including 3 harness failures fixed and disclosed, not hidden. Headline finding: steady noise (dishwasher 100%, hood fan 97%) degrades gracefully, but transient speech-like noise (fryer, a shouting coworker) hits accuracy far harder — a real, actionable finding for kitchen layout, not just a score.

BUILT FOR A REAL KITCHEN SHIFT

Hands-free by default, not just voice-in

WORKS IN A LOUD KITCHEN

- Push-to-talk mode for very loud stations
- Full-screen, one-glance Big Display mode
- Audio cues: a beep on every log, a sharper alert tone on amber/red

COVERS MORE OF THE REAL MENU

- Poultry, ground/injected meat, whole muscle, fish/seafood/eggs, reheating
- Two-stage cooling-curve tracking, not just point-in-time checks
- Every verdict cites its exact FDA Food Code section

Built for the people who actually read the log

THE PDF AN INSPECTOR EXPECTS

- Dedicated violations section: corrective action + the cook's exact spoken words as evidence
- Filterable by date and by station/item
- Page numbers, footer, manager sign-off line

A MANAGER CHECKING IN MID-SHIFT

- Live summary: readings logged, flagged, unresolved corrective actions
- Missed-check reminder if nothing's been logged in 45+ minutes
- One-click "Mark resolved" once a corrective action is actually taken

LIVE DEMO — VERIFIED AGAINST THE REAL DEPLOYED APP

Dashboard, mid-shift

38°F

Walk-in cooler — safe

43°F

Prep cooler — amber

152°F

Chicken breast — red, corrective action prompted

Every entry keeps the cook's exact spoken words next to the verdict, a manager can mark a corrective action resolved with one click, and the whole shift exports as an inspector-ready HACCP PDF with one click.

No mic? No problem — press “Try Demo”

- One click plays a ~30-second sample kitchen recording straight through the exact same real pipeline a live microphone uses.
- Nothing about the response is scripted: real AssemblyAI transcription, real rule-engine verdicts, real spoken confirmation, real UI updates from real tool calls.
- Demonstrates a safe reading, a flagged reading with a corrective action, a mid-sentence self-correction, and an amber boundary case — all in well under a minute.

BUSINESS VALUE & ORIGINALITY

Every kitchen already has to do this — almost nobody else is building it

BUSINESS VALUE

- Temperature logging is a health-code requirement, not optional, at every US food establishment.
- Turns a paper chore into a real-time, hands-free, provably-checked habit — the exported PDF is already in the format an inspector expects.
- No new hardware: runs in a browser tab a kitchen already has open.

ORIGINALITY

- Most of the gallery is general-purpose copilots, dispatch/fraud agents, or interview coaches.
- TempCheck is deliberately narrow: one real workflow, done with enough rigor — deterministic rules, mandatory readback, measured accuracy, real end-to-end verification, a judge-friendly demo mode — that it could plausibly go into a real kitchen tomorrow.

KNOWN LIMITATIONS, STATED PLAINLY

What isn't solved yet

- The accuracy test set is synthesized speech and noise, not real kitchen recordings — a disclosed, reproducible proxy, not a certified real-world number.
- Transient, speech-like noise (a fryer, a shouting coworker) is measurably harder for the STT than steady noise — 52.8%/47.2% vs. 88–100%.
- Cooling-curve tracking pairs a start and one later check and assumes temperature only decreased in between — it doesn't independently confirm an intermediate point, and the pending state doesn't survive a reload.
- Shift storage is local to one browser (localStorage) — no multi-device or multi-user backend yet.

LINKS

Try it yourself

- Live demo: tempcheck-vsh.vercel.app (click "Try Demo" — no mic needed)
- Code: github.com/pranuthi777/tempcheck-vs
- Measured accuracy & method: [docs/accuracy.md](https://docs/tempcheck-vs.com/accuracy.md)